

Credit Name: CSE 2110

Assignment Name: Chapter 5

Question #1

- a) To repeat a segment of code multiple times

Question #2

A while loop will run a program if the condition specified is met, a do while loop will run that loop once and the check before looping or moving on

Question #3

Review account setup

Question #4

- a) A loop that will go on indefinitely
- b) The condition that the loop requires is never changed within the program - eg: if while a boolean is true is the required condition, but its state is never updated within the program, it has nothing to end itself
- c) The value is too high for the specified condition

Question #5

The loop will run for 60 times

Question #6

No matter how high or low x is initially set to it will always reach a value that ends the loop

Question #7

- a) A counter is something that will change a variable by a fixed amount each time
 - i) $x = x + 1$
 - ii) Counters can be used for counting the number of times the loop has been run or keeping track of something like the number of students
- b) An accumulator does not need to change by a fixed amount
 - i) $X = x + y$
 - ii) Accumulators can be used for summing up the values of different variables like groceries or random number

Question #8

```
int x = 3;
int total = 0;
for (int i = 3; i <=10; i ++){
    total += x;
    x += 1;
}
```

Question #9

Does it need to run at least once (do-while), is a range(for) or a condition(while) needed

Question #11

String x = "my string"

- a) x.length();
 - i) Outputs 9
- b) x.substring(0,3);
 - i) 0-1 but 3 is not inclusive
 - ii) "my"
- c) x.toLowerCase();
 - i) All characters become lower case
 - ii) "my string"
- d) x.toUpperCase();
 - i) All character become uppercase
 - ii) "MY STRING"
- e) x.trim();
 - i) Trims extra spaces at the front and the back
 - ii) "my string"